Basanta Chaulagain

Detection Engineering | Behavioral Analytics | Digital Forensics | AI-assisted Security Automation

basanta.chaulagain@uga.edu | (706)-714-5079 | GitHub | LinkedIn | Website

EDUCATION

PhD in Computer Science | University of Georgia, Athens, GA **(GPA: 4.0)** Aug 2021 – July 2026

Bachelors in Computer Engineering | Tribhuvan University, Pulchowk Campus Nov 2014 – Sep 2018

SKILLS

- **Languages** Python (8+ years), C/C++ (6+ years), Bash (5+ years), PHP, Rust, Java. **Frameworks:** Django, Flask.
- **Data/Tools** Spark, Athena, SQL, JSON, REST API, Regex, eBPF, Docker, KVM, AWS. **Systems:** Linux, Windows.
- **AI/ML** AutoGen, LLM/agent, TensorFlow, PyTorch, Pandas, NLTK, Transformers. **Compliance**: NIST, SOC2.
- **Security** SIEM, UEBA, SOAR, Sigma, YARA, Burp Suite, Qualys, Wireshark, Cryptography, TCP/IP, NetworkX.

EXPERIENCE

Graduate Research Assistant Aug 2021 – Dec 2025

The University of Georgia Athens, GA

- Focused on **digital forensics**, software security, and **privacy-preserving techniques**.
- Led **end-to-end research lifecycle**: ideation, design, implementation, evaluation, and writing of multiple research projects.
- Designed and implemented **automated threat detection** and forensic analysis systems that correlate endpoint telemetry and security artifacts to **validate alerts** and improve detection accuracy across projects such as **CLEAR** and **FACT**.
- Analyzed **attacker techniques** such as persistence, privilege escalation, and defense evasion, and used those findings to design and evaluate detection methods.

Security Analytics Engineer Oct 2020 – Jul 2021

LogPoint (SIEM solution) Copenhagen, Denmark

- Built and tuned **100+ behavioral detection rules** and behavioral analytics mapped to **MITRE ATT&CK** framework to improve threat hunting, reduce false positives, and strengthen incident response workflows.
- Engineered **30+** security analytics applications for large-scale log ingestion, **threat detection**, anomaly identification, and **incident investigation** across diverse enterprise environments.

Software Engineer - Customer Success Oct 2018 – Oct 2020

LogPoint (SIEM solution) Copenhagen, Denmark

- Debugged product-related problems and developed hotfixes, often applying fixes in the **live production environment**.
- Developed and integrated **50+ plugins** for technical proof of concept, converting **70%** of demos to business.
- Defined KPI metrics and automated system monitoring, improving efficiency by **20%** and reducing manual effort by **30%**.
- Conducted **20+** certified user and administrator training sessions, contributing to product adoption by **120+** individuals.

PROJECTS/PUBLICATIONS

FA-SEAL: Forensically Analyzable Symmetric Encryption for Audit Logs **[ACSAC'24]**

- Designed and implemented a **privacy-preserving cryptographic** system that enables forensic analysis on encrypted audit logs with only **0.68% of log data disclosure**, processing **30GB** of high-volume daily logs in ~**90 minutes**.

CLEAR: Contextual Evidence-Based Alert Resolution for ML-Based HIDS **[Submitted to NDSS'27]**

- Built a **staged verification pipeline** for ML-based IDS alerts using live host metadata and artifacts, **resolving 97%+** of flagged entities across 18 attack scenarios with a mean verification time of **1.8s per entity**.

FACT: Forensic-Aware Artifact Collection System **[Submitted to ACSAC'26]**

- Developed an automated forensic-aware artifact collection system to support IDS **alert validation**, **incident investigation**, and threat verification, reducing **false positives to near zero** across 11 attack scenarios.

SynthDB: Synthesizing Database via Program Analysis for Security Testing of Web Applications **[NDSS'23]**

- Evaluated real-world PHP applications with SynthDB, a databases synthesizer for dynamic analysis of PHP web applications; and discovering **33 previously unknown vulnerabilities** from 5 real-world applications.

MAPSEC (Multi-Agentic Program for Security and Cyberdefense) **[MAPSEC]**

- Led development of an AI-assisted security operations prototype focused on **automating threat analysis** and improving security investigation workflows through **collaborative agents**.